

A Weather-Integrated Two-Stage XGBoost Framework for Subway Congestion Forecasting Under Large Events and Weather Threshold Analysis

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I. INTRODUCTION

The surge in subway congestion that occurs during large-scale events hinders efficient train operations [1]. In particular, while weather conditions are known to have complex effects on passenger travel behavior [2]–[4], the actual causal relationships in specific event scenarios have not been clearly established. Therefore, this study proposes a two-stage residual correction framework based on XGBoost that integrates weather data under large events. At the same time, it aims to elucidate the specific characteristics of weather’s influence on event congestion by analyzing trends in ridership indices based on precipitation and temperature at a representative station.

II. METHODOLOGY

This study proposes a two-stage residual correction model based on XGBoost [5] to account for the abrupt nonlinear congestion patterns observed during large-scale events. The first-stage model predicts the baseline ridership indices, while the second-stage model is designed to focus on learning the residuals between these predictions and the actual values, thereby correcting for event effects and improving the final prediction accuracy.

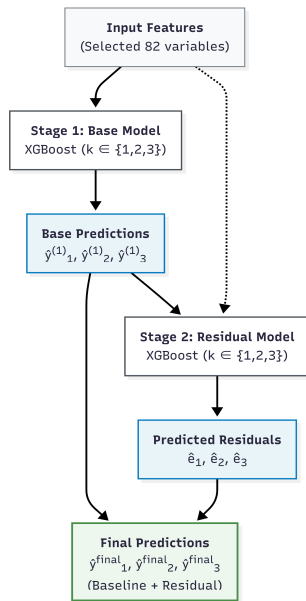


Fig. 1. Proposed two-stage residual correction framework for subway congestion forecasting under large events.

To formalize the proposed multi-target two-stage framework, the base XGBoost model in the first stage predicts the baseline $\hat{y}_{i,k}^{(1)}$ by minimizing $\mathcal{L}_{1,k}$:

$$\mathcal{L}_{1,k} = \sum_{i=1}^n l(y_{i,k}, \hat{y}_{i,k}^{(1)}) + \Omega(f_k^{(1)}), \quad k \in \{1, 2, 3\} \quad (1)$$

Then, the residual $e_{i,k} = y_{i,k} - \hat{y}_{i,k}^{(1)}$ is extracted, and the second-stage residual-based XGBoost model minimizes $\mathcal{L}_{2,k}$ to predict these errors ($\hat{e}_{i,k}$):

$$\mathcal{L}_{2,k} = \sum_{i=1}^n l(e_{i,k}, \hat{e}_{i,k}) + \Omega(f_k^{(2)}) \quad (2)$$

The final event-corrected prediction is formulated as $\hat{y}_{i,k}^{final} = \hat{y}_{i,k}^{(1)} + \hat{e}_{i,k}$, where i , n , k , and Ω denote the sample index, total observations, target variables, and regularization term, respectively.

In this study, we constructed a multi-source dataset with 10-minute intervals covering the 2024–2025 period and divided it into training (2024), validation (March–April 2025), and test (May–July 2025) datasets.

- 1) *Subway Ridership and Event Data*: Seoul Metro’s station-specific ridership indices (on-board, boarding, and alighting) collected at 10-minute intervals and train operation data were linked with provided event information (type, start and end times).
- 2) *Real-time and Forecast Weather Data*: The Korea Meteorological Administration’s 10-minute interval real-time weather data and special weather advisory data were utilized. To simulate actual operating conditions during the testing phase, hourly short-term forecast data were interpolated into 10-minute intervals to match the temporal resolution of the other sources.

To capture nonlinear congestion characteristics, lagged ridership indices were incorporated alongside weather and event data to train the proposed framework for the six target stations.

III. RESULTS

A. Model Performance

Table I presents the performance comparison focusing on the Inner Circle of Line 2 Sports Complex Station as a representative case. The proposed model effectively captured the nonlinear congestion patterns

TABLE I
PERFORMANCE COMPARISON OF PREDICTION MODELS FOR LINE
2 SPORTS COMPLEX STATION (INNER CIRCLE) DURING
WEEKDAYS AND WEEKENDS.

Target Value	Weekday			Weekend		
	R ²	RMSE	MAPE	R ²	RMSE	MAPE
On board	0.96	288.60	16.00	0.84	191.25	0.81
Boarding	0.87	85.42	27.96	0.70	102.65	66.03
Alighting	0.90	52.05	32.54	0.81	48.88	30.47

at Sports Complex Station (Lines 2 and 9) on weekdays. On weekdays, where patterns are well-defined, the model demonstrated good accuracy ($R^2 \geq 0.77$) across all three indices; however, on weekends, prediction performance declined moderately ($R^2 \geq 0.61$) due to the high data skewness caused by overlapping events and partial omissions in the provided event logs. Meanwhile, World Cup Stadium Station (Line 6), characterized by low event frequency and limited samples, recorded the lowest performance, underscoring the necessity of a critical mass of training data to ensure learning stability.

B. Correlation Analysis of Precipitation Levels and Congestion during Event-Affected Hours

TABLE II
PRECIPITATION CORRELATION RESULTS DURING
EVENT-AFFECTED HOURS

Day	Time	Comparison Pair	Result (p)
Wd.	20:00	— (Lack of Data)	—
	22:00	No Rain vs. Light/Mod.	> 0.05
		Strong / Very Strong	(Lack of Data)
	23:00	No Rain vs. Mod.	> 0.05
		Light / Strong / Very Strong	(Lack of Data)
Wk.	15:00	— (Lack of Data)	—
	17:00	No Rain vs. Light	> 0.05
		Mod. / Strong / Very Strong	(Lack of Data)
	18:00	No Rain vs. Mod.	> 0.05
		Light / Strong / Very Strong	(Lack of Data)

To control for spatial characteristics, we analyzed differences in ridership indices across five precipitation levels at Line 2 Sports Complex Station (Inner Circle), which possesses the most extensive event data. To account for data non-normality and sample size imbalances, the non-parametric Mann-Whitney U test was employed, integrated with bootstrapping ($n = 1,000$) to enhance statistical robustness.

The analysis revealed no statistically significant differences on either weekdays or weekends. Detailed analysis by time slot also showed no observable changes in ridership patterns under light or moderate rain conditions compared to the “no rain” baseline. This suggests that, under the context of travel with the clear purpose of attending a large-scale event, general levels of precipitation are not a dominant factor influencing subway ridership decisions.

C. Correlation Analysis of Temperature and Congestion During Event-Affected Hours

A nonparametric LOWESS trend analysis of the historical data for the same representative station revealed no distinct variations in ridership indices

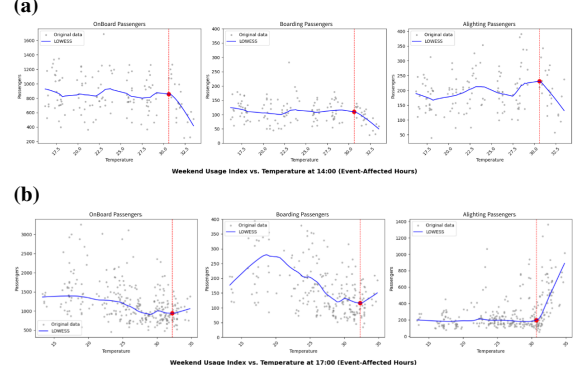


Fig. 2. Changes in subway ridership indices by temperature during weekend event-affected hours: (a) 14:00 and (b) 17:00.

in response to temperature changes during weekday event-affected hours (7:00–10:00 PM). However, the results during the weekend event revealed varying threshold effects by time of day, with temperatures around 30°C serving as a key point. During the daytime (2:00–3:00 PM), a ridership-suppressing effect prevailed at temperatures above 30°C, causing the ridership indices to drop sharply; conversely, during the late afternoon (5:00–6:00 PM), the ridership indices actually increased despite the high temperatures. This demonstrates that time pressure exerted a more dominant influence on travel decisions than thermal discomfort in the late afternoon.

IV. IMPACTS

This framework improves transit operation efficiency and user convenience by predicting congestion during large-scale events in advance. In particular, by reflecting the distinct threshold characteristics of daytime hours—when ridership is suppressed due to high temperatures exceeding 30°C—and the afternoon hours—when crowd density surges due to event schedules despite such high temperatures—it enables customized station environment control and proactive train scheduling. This is significant in that it provides guidelines for flexibly responding to complex urban transit congestion situations where weather conditions and crowd density intersect.

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